

Nonconvex Fisher–Rao Instability and Projected KL Stationarity

Abstract

We study deterministic one-dimensional Gaussian variational inference for the smooth but non-log-concave target

$$V_R(x) = \frac{1}{2}x^2 - 2R^2 \log \cosh(x/R), \quad -1 \leq V_R''(x) \leq 1.$$

Two complementary points are made. First, a globally bounded Hessian alone does *not* numerically control the trajectory-wise Fisher–Rao gradient: the Riemannian-exponential covariance step undergoes a negative-curvature cascade from $c_0 = 1$, and the KL/Bregman covariance step has a rational pole near the loss of positive definiteness. Second, when the covariance is constrained to a prescribed spectral interval $[\lambda_-, \lambda_+]$, the *projected* KL scheme of Section 2.4 (Theorem 2.18) of the manuscript has the correct constrained Bregman stationarity diagnostic: the running minimum of the Bregman displacement $\text{KL}(\rho_{a_n} \parallel \rho_{a_{n+1}})$ satisfies the energy-drop envelope $\min_{0 \leq n < N} \text{KL}(\rho_{a_n} \parallel \rho_{a_{n+1}}) \leq \frac{\Delta t}{N} \{\text{KL}(\rho_{a_0} \parallel \rho_{\text{post}}) - \text{KL}(\rho_{a_N} \parallel \rho_{\text{post}})\}$ in the theorem-safe stepsize regime. A small additional study probes the larger stepsize $\Delta t = 1/(\beta \lambda_+)$, which violates the theorem’s relative-smoothness condition ($\Delta t L_{\text{clip}} = 64 \gg 1$): the envelope still holds empirically, because the projected trajectory reaches the active upper constraint — a constrained-stationary point — in a single step. We emphasize what is and is not shown: the clipped scheme certifies constrained Bregman stationarity over the covariance-truncated feasible set; it does not make the unconstrained Fisher–Rao gradient small, nor does it converge to the unconstrained Gaussian variational optimum.

1 Setup

We target the one-dimensional posterior $\rho_{\text{post}} \propto e^{-V_R}$ with

$$V_R(x) = \frac{1}{2}x^2 - 2R^2 \log \cosh(x/R), \quad V_R'(x) = x - 2R \tanh(x/R), \quad V_R''(x) = 1 - 2\{1 - \tanh^2(x/R)\}.$$

The Hessian satisfies $-1 \leq V_R''(x) \leq 1$, with $V_R''(0) = -1$ and $V_R''(x) \rightarrow 1$ in the tails; the target is therefore globally smooth ($\|\nabla^2 V_R\| \leq \beta$ with $\beta = 1$) but non-log-concave. For $X \sim \mathcal{N}(0, c)$ define

$$A_R(c) = \mathbb{E}[V_R''(X)], \quad F_R(c) = \mathbb{E}[V_R(X)] - \frac{1}{2} \log c.$$

All trajectory experiments are symmetric ($m = 0$) and deterministic: the expectations $A_R(c)$ and $F_R(c)$ are evaluated by fixed Gauss–Hermite quadrature, and no normalizing constant for ρ_{post} is required. The function F_R is the Gaussian variational objective $\text{KL}(\rho_a \parallel \rho_{\text{post}})$ up to the (unknown) additive log-partition constant; that constant cancels in every energy *drop* used below, so we always work with $F_R(c_0) - F_R(c_N)$.

2 Unprojected Fisher–Rao diagnostics

We record, along each trajectory, the squared Fisher–Rao and Bures–Wasserstein gradient norms of the symmetric Gaussian objective,

$$\|\nabla_{\text{FR}} F_R(c)\|^2 = (1 - cA_R(c))^2, \quad \|\nabla_{\text{BW}} F_R(c)\|^2 = \frac{(1 - cA_R(c))^2}{c}.$$

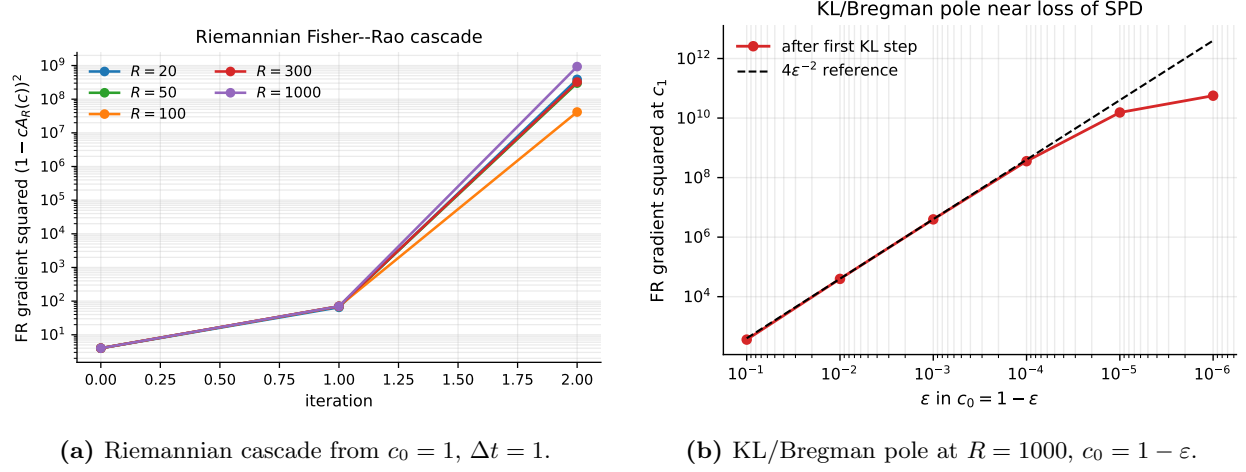


Figure 1: Unprojected Fisher–Rao diagnostics. The trajectory-wise FR gradient squared is not controlled by the global Hessian bound $|V_R''| \leq 1$: the Riemannian step amplifies $(1 - cA_R(c))^2$ through a negative-curvature cascade (left), and the KL step exhibits a $\propto \varepsilon^{-2}$ pole as the denominator $1 + \Delta t c_0 A_R(c_0)$ approaches zero (right).

Both vanish at a Gaussian stationary point $cA_R(c) = 1$ and both are *covariance-coupled* through the factor $cA_R(c)$.

2.1 Riemannian covariance cascade

The Riemannian (geodesic) Fisher–Rao covariance step is

$$c_{n+1} = c_n \exp\{\Delta t (1 - c_n A_R(c_n))\},$$

implemented in the log domain with an overflow guard and a stopping rule $c_n > R^2/4$. From $c_0 = 1$ and large R the Gaussian cloud sees almost constant negative curvature $A_R(c) \approx -1$, so the multiplicative factor is $\approx e^{1+c}$ and the covariance bursts upward. Figure 1a shows the resulting trajectory-wise FR gradient squared growing by several orders of magnitude despite $|V_R''| \leq 1$.

2.2 KL/Bregman pole

The (unprojected) KL/Bregman Fisher–Rao covariance step is

$$c_{n+1} = \frac{(1 + \Delta t)c_n}{1 + \Delta t c_n A_R(c_n)}.$$

With $c_0 = 1 - \varepsilon$ and $A_R(c_0) \approx -1$ the denominator is $\approx \varepsilon$, so $c_1 \approx 2/\varepsilon$ and the next-state FR gradient follows a $\propto \varepsilon^{-2}$ pole until the finite- R curvature average stops behaving like a constant -1 . Figure 1b isolates this rational pole near the loss of positive definiteness.

3 Bures–Wasserstein running-minimum comparison

The Bures–Wasserstein forward–backward step is

$$\tilde{c}_n = (1 - \eta A_R(c_n))^2 c_n, \quad c_{n+1} = \frac{1}{2} \{\tilde{c}_n + 2\eta + \sqrt{\tilde{c}_n(\tilde{c}_n + 4\eta)}\},$$

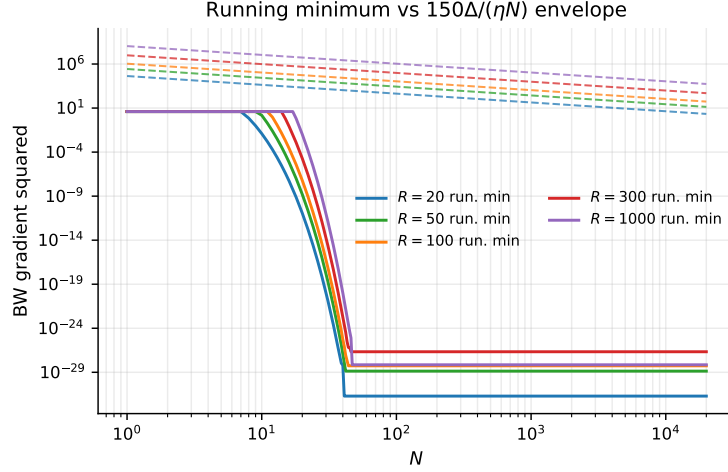


Figure 2: Bures–Wasserstein forward–backward runs, $\eta = 0.9$. Solid: running minimum of the BW gradient squared; dashed: the loose $150\Delta/(\eta N)$ envelope.

with $\eta = 0.9$. The relevant nonconvex stationarity quantity here is the *running minimum* of the BW gradient norm, $\min_{0 \leq k < N} \|\nabla_{\text{BW}} F_R(c_k)\|^2$, compared against an $O(1/N)$ envelope $150\Delta/(\eta N)$ with $\Delta = F_R(c_0) - F_{R,\star}^{\text{num}}$ (the envelope is deliberately loose because Δ scales with the target depth). Figure 2 confirms that the running minimum stays below the envelope across R . This comparison is geometrically distinct from Section 2: it is a running-minimum statement in the BW geometry, not a pointwise Fisher–Rao bound.

4 Projected KL scheme with covariance clipping

Sections 2–3 motivate *constraining the covariance*. Following Section 2.4 of the manuscript, fix $0 < \lambda_- \leq \lambda_+ < \infty$ and the covariance-truncated feasible set $\mathcal{A}' = \{a = (m, C) : \lambda_- I \preceq C \preceq \lambda_+ I\}$. Under the one-sided smoothness Assumption 2.17, $\|\nabla^2 V\| \leq \beta$, the projected KL scheme clips the unprojected KL update back into $[\lambda_-, \lambda_+]$. In the symmetric scalar case ($m = 0$, $\nabla^2 \log \rho_{\text{post}} = -V_R''$ so $\mathbb{E}[\nabla^2 \log \rho_{\text{post}}] = -A_R(c)$) the update is

$$\tilde{c}_{n+1} = \frac{(1 + \Delta t)c_n}{1 + \Delta t c_n A_R(c_n)}, \quad c_{n+1} = \text{clip}_{[\lambda_-, \lambda_+]}(\tilde{c}_{n+1}),$$

where $\text{clip}_{[\lambda_-, \lambda_+]}(x) = \min\{\lambda_+, \max\{\lambda_-, x\}\}$ is the one-dimensional eigenvalue clip.

Bregman displacement and energy-drop envelope. The one-step Bregman displacement is the KL divergence between successive iterates, $D_n = \text{KL}(\rho_{a_n} \parallel \rho_{a_{n+1}})$, which for the scalar symmetric Gaussians reduces to

$$D_n = \text{KL}(\mathcal{N}(0, c_n) \parallel \mathcal{N}(0, c_{n+1})) = \frac{1}{2} \left(\frac{c_n}{c_{n+1}} - 1 - \log \frac{c_n}{c_{n+1}} \right) \geq 0,$$

with equality iff $c_n = c_{n+1}$. For each prefix horizon N we form the running minimum $D_{\min}(N) = \min_{0 \leq n < N} D_n$ and the energy-drop envelope

$$B_N = \frac{\Delta t}{N} \{F_R(c_0) - F_R(c_N)\}.$$

R	Δt	$D_{\min}(N)$	B_N	max. viol.	check
20	7×10^{-3}	0	2×10^{-5}	0	yes
50	7×10^{-3}	0	2×10^{-5}	0	yes
100	7×10^{-3}	0	2×10^{-5}	0	yes
300	7×10^{-3}	0	2×10^{-5}	0	yes
1000	7×10^{-3}	0	2×10^{-5}	0	yes

Table 1: Projected-KL stationarity summary by R : stepsize Δt , final running minimum $D_{\min}(N)$, final envelope B_N , the maximum violation $\max_N \max\{0, D_{\min}(N) - B_N\}$, and the theorem check. The check passes for every R with maximum violation 0.

Theorem 2.18 gives the constrained Bregman stationarity guarantee

$$\min_{0 \leq n \leq N-1} \text{KL}(\rho_{a_n} \| \rho_{a_{n+1}}) \leq \frac{\Delta t}{N} \{ \text{KL}(\rho_{a_0} \| \rho_{\text{post}}) - \text{KL}(\rho_{a_N} \| \rho_{\text{post}}) \}, \quad \text{i.e.} \quad D_{\min}(N) \leq B_N,$$

valid for the theorem-safe stepsize. Since $\beta = 1$, the Bregman smoothness constant and stepsize are

$$L_{\text{clip}} = \beta \max \left\{ \lambda_+, \frac{\lambda_+^4}{\lambda_-^3} \right\}, \quad \Delta t = \frac{\text{safety}}{L_{\text{clip}}}, \quad \text{safety} < 1.$$

With $\lambda_- = 0.5$, $\lambda_+ = 2.0$, and $\text{safety} = 0.9$ this gives $L_{\text{clip}} = \max\{2, 16/0.125\} = 128$ and $\Delta t = 0.9/128 \approx 7.0 \times 10^{-3}$. In this regime the denominator $1 + \Delta t c_n A_R(c_n) \geq 1 - \Delta t \beta \lambda_+ \geq 1 - \text{safety} > 0$ is positive for every feasible $c_n \in [\lambda_-, \lambda_+]$ and every $A_R \in [-\beta, \beta]$, so the pole of Section 2.2 cannot occur.

A non-theorem-safe stepsize. The Theorem 2.18 stepsize is sufficient but conservative, and it is natural to ask whether the stationarity envelope still follows when it is violated. We therefore also run the projected scheme at the larger Riemannian-scale stepsize

$$\Delta t = \frac{1}{\beta \lambda_+} = 0.5, \quad \text{so} \quad \Delta t L_{\text{clip}} = \frac{\lambda_+^4}{\lambda_-^3} \cdot \frac{1}{\lambda_+} = \frac{\lambda_+^3}{\lambda_-^3} = 64 \gg 1,$$

which lies well outside the relative-smoothness condition $\Delta t \leq 1/L_{\text{clip}}$ that underlies the one-step descent inequality, so Theorem 2.18 gives no guarantee. The value $1/(\beta \lambda_+)$ is the largest stepsize for which the denominator bound $1 + \Delta t c_n A_R(c_n) \geq 1 - \Delta t \beta \lambda_+ = 0$ is still (weakly) nonnegative over the feasible box; beyond it the unprojected KL covariance step can hit the genuine sign change of Section 2.2 that the clip cannot mask.

5 Results and interpretation

Unprojected results motivate covariance control. Table 2 and Figure 1 show that, without any covariance constraint, both Fisher–Rao discretizations amplify the trajectory-wise gradient factor $(1 - c A_R(c))^2$ on this globally smooth target: the Riemannian step by 4–5 orders of magnitude before its stopping rule, and the KL step along a $\propto \varepsilon^{-2}$ pole. The obstruction is not roughness of V_R ; it is the coupling of negative curvature to the covariance.

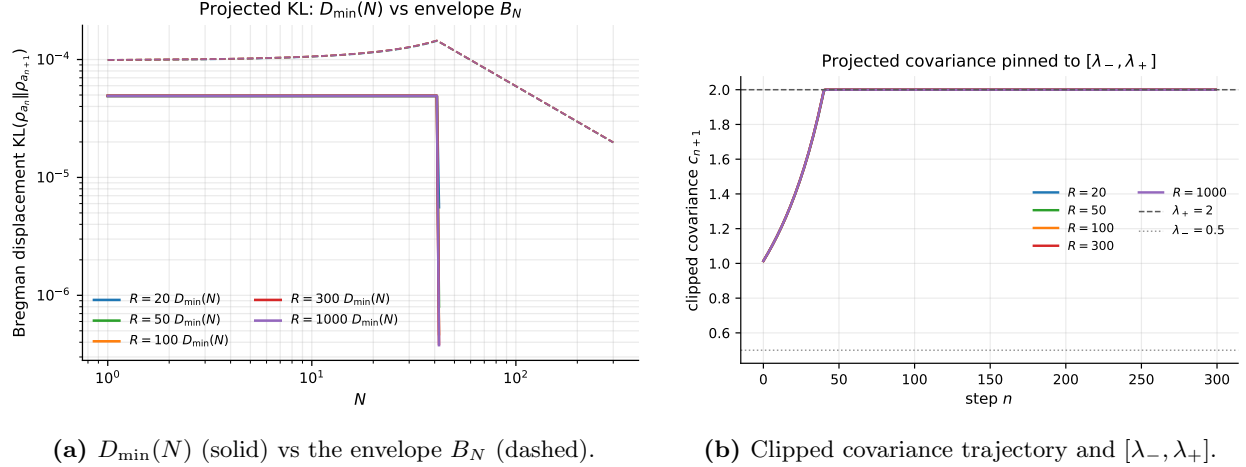


Figure 3: Projected KL scheme, $\lambda_- = 0.5$, $\lambda_+ = 2.0$, $\beta = 1$, $\Delta t = 0.9/128$. The running-minimum Bregman displacement stays below the energy-drop envelope at every prefix; once the upper constraint binds (step ≈ 41) the projected step is stationary at the boundary, the covariance is pinned at λ_+ , and D_n (hence $D_{\min}$) collapses to zero.

Clipping removes the pole in the theorem-safe regime. Restricting the covariance to $[\lambda_-, \lambda_+]$ and using the theorem-safe $\Delta t = 0.9/L_{\text{clip}}$ keeps the denominator bounded away from zero, so the projected KL trajectory is well defined for all N (Figure 3b). From $c_0 = 1$ the negative curvature drives the covariance monotonically up to the upper bound $\lambda_+ = 2$, where it is pinned; the covariance never approaches the SPD boundary that produced the pole.

The running-minimum Bregman displacement satisfies the envelope. Figure 3a and Table 1 verify the Theorem 2.18 check: at every prefix horizon N the running minimum $D_{\min}(N)$ lies below the energy-drop envelope B_N , with maximum violation 0 across all R . The energy $F_R(c_n)$ is monotonically nonincreasing along the trajectory (so $B_N \geq 0$), and the diagnostic is essentially R -independent because the dynamics in $c \in [1, 2]$ see $A_R(c) \approx -1$ for every R in the grid.

Boundary stationarity is constrained, not unconstrained, smallness. Once the upper covariance constraint becomes active, the clipped update returns $c_{n+1} = \lambda_+ = c_n$, so $D_n = \text{KL}(\rho_{a_n} \parallel \rho_{a_{n+1}}) = 0$ exactly and $D_{\min}(N)$ collapses to zero. This is *constrained Bregman stationarity*: the projected step is stationary on the boundary of the feasible set $\mathcal{A}'$. It is *not* a statement that the unconstrained Fisher–Rao gradient $(1 - cA_R(c))^2$ is small there — at $c = \lambda_+ = 2$ with $A_R \approx -1$ the unconstrained FR gradient is in fact $O(1)$ and points outward, which is precisely why the iterate stays clipped. The correct reading of the experiment is therefore stationarity over the covariance-truncated feasible set, not unconstrained optimality.

5.1 Beyond the theorem-safe regime

We repeat the projected scheme at the larger stepsize $\Delta t = 1/(\beta\lambda_+) = 0.5$ introduced at the end of Section 4, for which $\Delta t L_{\text{clip}} = 64 \gg 1$ and Theorem 2.18 offers no guarantee. Figure 4 and Table 3 report the outcome. *Empirically the stationarity envelope $D_{\min}(N) \leq B_N$ still holds, with maximum violation 0 for every R .* The mechanism is visible in the figure: the single large step makes a comparatively large energy drop and clips the covariance onto the upper bound λ_+ in one move (Figure 4, right), so the only nonzero displacement is the transient D_0 , which is comfortably below

experiment	R	max / slope	terminal value	check
Riemannian cascade	20	2×10^4	2.4×10^4	c_threshold
Riemannian cascade	50	1.7×10^4	3.1×10^4	c_threshold
Riemannian cascade	100	6.4×10^3	3.2×10^4	c_threshold
Riemannian cascade	300	1.8×10^4	3.2×10^4	c_threshold
Riemannian cascade	1000	3.1×10^4	3.2×10^4	log_c_overflow_guard
KL pole	1000	-1.998	5.6×10^{10}	pole-window slope
BW forward-backward	20	2×10^{-32}	2.17	yes
BW forward-backward	50	1.4×10^{-29}	13.6	yes
BW forward-backward	100	5.4×10^{-29}	54.4	yes
BW forward-backward	300	2.1×10^{-27}	490	yes
BW forward-backward	1000	7.1×10^{-29}	5.4×10^3	yes

Table 2: Unprojected diagnostics. Riemannian rows report the maximum FR gradient norm along each cascade and the stop reason; the KL row reports the log-log pole-window slope; the BW rows report the final running minimum and envelope check.

the first-prefix envelope B_1 , and from the second step onward the active-constraint stationarity gives $D_n = 0$.

We are careful about what this does and does not show. It is *not* evidence that the relative-smoothness descent inequality $\mathbb{E}(a_{n+1}) \leq \mathbb{E}(a_n) - \Delta t^{-1} D(a_n, a_{n+1})$ holds at this stepsize in general; rather, on this target the projected trajectory reaches a constrained-stationary boundary point in a single step, after which the running-minimum check is satisfied trivially. Two features make the larger stepsize benign here and warn against extrapolating: the energy still decreases monotonically (so $B_N \geq 0$), and the denominator $1 + \Delta t c_n A_R(c_n)$ stays positive but grazes $\approx 4 \times 10^{-6}$ at $c_n = \lambda_+$ for large R — exactly the sign change of Section 2.2 that the choice $\Delta t = 1/(\beta\lambda_+)$ sits on the edge of and that the clip is masking. A larger stepsize, or a target whose curvature average reaches $-\beta$ inside the feasible box, would cross that edge. The conclusion is therefore narrow: the theorem stepsize is sufficient but not necessary for the stationarity envelope on this example, and the constraint set, not the stepsize bound, is what stabilizes the trajectory.

6 Limitations

The evidence is deterministic, one-dimensional, finite-precision, and finite-horizon. The unprojected experiments support the *absence* of a useful smoothness-only, covariance-free, trajectory-wise Fisher–Rao bound on this target, but a finite numerical study is not a formal impossibility theorem. For the projected scheme we make only the claims the theory supports: in the theorem-safe stepsize regime the projected KL iterates satisfy the constrained Bregman stationarity envelope $D_{\min}(N) \leq B_N$, and the active-constraint plateau is constrained stationarity on $\partial\mathcal{A}'$. We do *not* claim that clipping makes the unconstrained Fisher–Rao gradient small, nor that the projected scheme converges to the unconstrained Gaussian variational optimum a_* ; indeed the optimum need not lie in $[\lambda_-, \lambda_+]$, and here the iterate is pinned at λ_+ . The non-theorem-safe study of Section 5.1 is a single-example observation, not a relaxation of the theorem’s hypotheses: the envelope holds there because the trajectory reaches a constrained-stationary boundary point in one step, while the denominator sits on the edge of the sign change that the clip is masking, so the larger stepsize should not be expected to be safe in general. The Gauss–Hermite quadrature order, the clipping bounds, the stepsize-safety

Projected KL beyond the theorem-safe regime: $\Delta t = 1/(\beta\lambda_+)$, $\Delta t L_{\text{clip}} = 64$

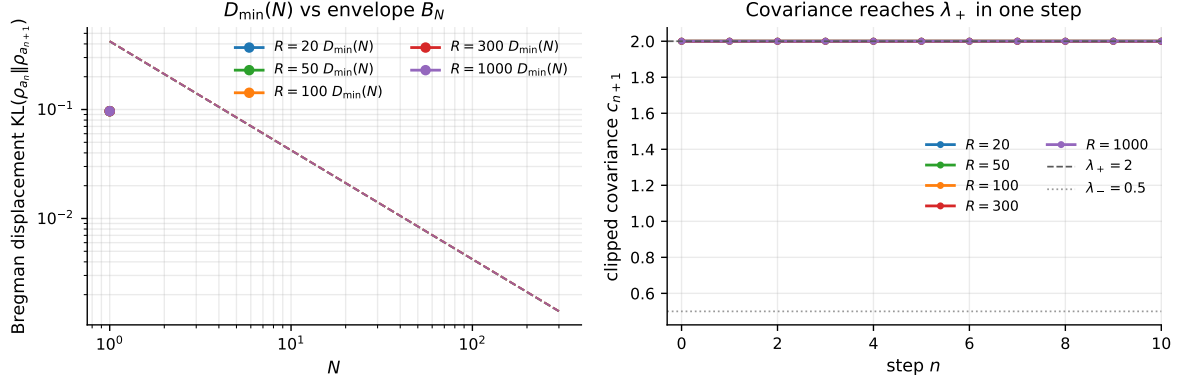


Figure 4: Projected KL at the non-theorem-safe stepsize $\Delta t = 1/(\beta\lambda_+) = 0.5$ ($\Delta t L_{\text{clip}} = 64$). Left: the running minimum $D_{\min}(N)$ (markers) versus the envelope B_N (dashed); the only nonzero value is the transient D_0 at $N = 1$, below B_1 , after which $D_{\min}$ collapses to zero. Right: the covariance reaches λ_+ in a single step and is pinned there.

factor, the comparison stepsize $1/(\beta\lambda_+)$, and the smoothness constant $\beta = 1$ are recorded with the output metadata for reproducibility.

R	regime	Δt	$\Delta t L_{\text{clip}}$	$D_{\min}(N)$	max. viol.	energy mono.	check
20	theorem-safe	7×10^{-3}	0.9	0	0	yes	yes
	$1/(\beta\lambda_+)$	0.5	64	0	0	yes	yes
50	theorem-safe	7×10^{-3}	0.9	0	0	yes	yes
	$1/(\beta\lambda_+)$	0.5	64	0	0	yes	yes
100	theorem-safe	7×10^{-3}	0.9	0	0	yes	yes
	$1/(\beta\lambda_+)$	0.5	64	0	0	yes	yes
300	theorem-safe	7×10^{-3}	0.9	0	0	yes	yes
	$1/(\beta\lambda_+)$	0.5	64	0	0	yes	yes
1000	theorem-safe	7×10^{-3}	0.9	0	0	yes	yes
	$1/(\beta\lambda_+)$	0.5	64	0	0	yes	yes

Table 3: Theorem-safe versus non-theorem-safe stepsize, by R : stepsize Δt , the relative-smoothness ratio $\Delta t L_{\text{clip}}$ (< 1 is theorem-safe), the final running minimum $D_{\min}(N)$, the maximum violation, energy monotonicity, and the envelope check. The envelope holds in both regimes even though the larger stepsize has $\Delta t L_{\text{clip}} = 64 \gg 1$.